Computers in Human Behavior 144 (2023) 107723

Contents lists available at ScienceDirect

Computers in Human Behavior

journal homepage: www.elsevier.com/locate/comphumbeh

Can I @handle it? The effects of sponsorship disclosure in TikTok influencer marketing videos with different product integration levels on adolescents' persuasion knowledge and brand outcomes

Daphne L.M. van der Bend [a,b,*], Nerine Gijsman [a], Tamara Bucher [c], Vanessa A. Shrewsbury [b], Hans van Trijp [a], Ellen van Kleef [a]

[a] Department of Social Sciences, Marketing and Consumer Behaviour Group, Wageningen University & Research, Hollandseweg 1, 6706, KN, Wageningen, the Netherlands
[b] School of Health Sciences, College of Health, Medicine and Wellbeing, University of Newcastle, University Drive, Callaghan, NSW, 2308, Newcastle, Australia
[c] School of Environmental and Life Sciences, College of Engineering, Science and Environment, The University of Newcastle, University Drive, Callaghan, NSW, 2308, Newcastle, Australia

ARTICLE INFO

Handling Editor: Nicolae Nistor

Keywords:
Influencer marketing
Social media
TikTok
Persuasion knowledge
Adolescents
Brand outcomes

ABSTRACT

TikTok influencer videos are becoming increasingly popular for promotional purposes and are appealing to adolescents. Yet, the products' level of integration into the storyline of such videos influences their recognizability as advertisements, which is concerning as they expose adolescents to unhealthy social media food marketing, leading to unhealthy consumption. This study investigated whether a sponsorship disclosure in a TikTok influencer video with different levels of product-plot integration increases adolescents' persuasion knowledge and impacts their brand outcomes. This between-subjects 2 × 2 experiment included 245 Dutch adolescents with a mean age of 13.6 (SD = 1.42). Participants watched a TikTok video in which the brand Doritos was shown passively or integrated into the storyline, with or without a disclosure. The sponsorship disclosure significantly increased recognition of advertising ($F_{(241)} = 4.48$, $p = .035$) and understanding of persuasive intent ($F_{(241)} = 13.28$, $p = <.001$), regardless of product-plot integration level. Yet, brand attitude and product choice were not affected significantly by the disclosure. Overall, sponsorship disclosure in TikTok influencer videos helps create transparency and increase adolescents' recognition and understanding of commercial and persuasive intent. This study is the first to investigate disclosures in the context of popular short-format social media videos.

1. Introduction

Adolescents spend an increasing amount of time on online platforms. Almost 99% of Dutch adolescents aged 12–18 years uses a smartphone to visit the Internet (CBS, 2020). The majority of the time adolescents use electronic devices is spent on social media platforms in particular, exposing them to new forms of marketing content that is blurred with entertainment (Lapierre, Fleming-Milici, Rozendaal, McAlister, & Castonguay, 2017). On social media, marketers apply rather stealth product placements by subtly placing their products into a storyline that distracts adolescents from the commercial intent, for example by using their personal data to connect to their interest and encouraging them to actively engage with the content (Boyland and Tatlow-Golden, 2017; World Health Organization, 2016). Another frequently used product placement technique on social media involves social media influencers who promote products or brands in video blogs (vlogs). Social media product placements are considered fundamentally different from television product placements and much more powerful, as the involvement of influencers has changed the contexts and storylines products are placed in (Boyland & Tatlow-Golden, 2017; De Veirman, Hudders, & Nelson, 2019). Influencers can incorporate product placement in various ways, from subtle background placements to prominent integration into the video's storyline (such as using or consuming the product). Adolescents' reactions and perceptions of the marketed product may be influenced by the level of product integration utilized by the influencer (Valenzuela, Martinez-Troncoso, & Yanez-Wieland, 2015; Yang &

* Corresponding author. Department of Social Sciences, Marketing and Consumer Behaviour Group, Wageningen University & Research, Hollandseweg 1, 6706, KN, Wageningen, the Netherlands.
E-mail address: daphne.vanderbend@wur.nl (D.L.M. van der Bend).

https://doi.org/10.1016/j.chb.2023.107723
Received 16 November 2022; Received in revised form 26 February 2023; Accepted 28 February 2023
Available online 6 March 2023
0747-5632/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Roskos-Ewoldsen, 2007). Overall, adolescents may be less resistant and more trusting towards brand promotions from influencers than marketing communications from brands (De Veirman et al., 2019).

When influencers are intentionally advertising products in return for free promotional goods or payment, this content is referred to as influencer marketing (De Veirman et al., 2019). Marketers have started to increasingly employ social media influencers in their campaigns. In fact, the influencer marketing industry is set to grow to about $16,4 billion in 2022 (Geyser, 2022). Nowadays any social media user can make money by (re-)creating content and contributing to the marketing of brands. Increased popularity and engagement with their posts may subsequently encourage them to initiate deals with brands that in turn help them post more creative content on their social media accounts (De Veirman, De Jans, Van den Abeele, & Hudders, 2020, pp. 126–166). Such nano-influencers typically have 1000 to 10,000 followers and are much more relatable for adolescents, as compared to world famous mega- or macro-influencers with over 100,000 followers whose lives are much less attainable or realistic (Gillespie, 2021). Since social media users view their relationship with influencers as friendship rather than fan ship (O'Neil-Hart & Blumenstein, 2016), influencers may play an important role in brand building and increasing the effectiveness of social media brand promotions (Lou & Kim, 2019). Consequently, brands have now started hiring groups of nano-influencers for their campaigns instead of just one mega- or macro-influencer (Gillespie, 2021). Particularly entertaining, short-format influencer videos that have become popular with the rise of social media video platform TikTok and can now also be viewed on platforms such as YouTube, Instagram and Snapchat, are popular for promotional purposes and are highly appreciated and viewed by adolescents (van der Bend et al., 2022a, 2022c). TikTok is the fastest growing social media platform in the Netherlands, according to a 2022 report (Newcom, 2022). However, commercial TikTok influencer videos often blend in with regular entertaining content, making it difficult to recognize them as advertising. This may lower adolescents' awareness of and critical beliefs towards the persuasive intent of these videos (De Jans & Hudders, 2020; Ofcom, 2017, van der Bend et al., 2022b).

Sponsorship disclosures can make the commercial intent of influencer marketing content more transparent and help adolescents recognize and understand the commercial intent of TikTok influencer marketing videos (De Jans, Cauberghe, & Hudders, 2018). Yet, current regulations on the use of disclosures by influencers do not suffice as in practice many social media influencers do still not appropriately disclose commercial intent (Van Der Goot, Van Reijmersdal, & Zandbergen, 2021). Specifically, in the Netherlands stricter legislative rules regarding influencer marketing that are being closely supervised by the Dutch Media Authority were only implemented recently (i.e., July 1, 2022), and cover paid marketing communications by influencers with more than 500,000 followers on TikTok. Thus, the Dutch Media Act does not include influencers with less followers and therefore these smaller influencers rely on self-regulatory guidelines and are not watched as closely (DDMA, 2022). To stimulate policy changes to help adolescents become aware of influencer marketing on TikTok, more research is needed on the effectiveness of sponsorship disclosure, in videos with different product placement levels.

This may be particularly important with regard to food marketing, as this has been argued to have a significant effect on adolescent health. Adolescents are exposed to large amounts of social media food marketing and food content nowadays, including on TikTok (CBS, 2020; van der Bend et al., 2022b; Potvin Kent, Pauzé, Roy, de Billy, & Czoli, 2019). Food promotions on social media were found to be highly appealing to adolescents, especially when they are entertaining, aesthetic, short format videos, and promote energy-dense nutrient-poor (EDNP) foods (van der Bend et al., 2022a, 2022c). Concerningly, the majority of food promotions on social media targeting adolescents were evaluated as EDNP, and this may have the effect of stimulating adolescents' consumption of such foods, hence increasing their risk of obesity (Dunlop, Freeman, & Jones, 2016; Kelly, Bosward, & Freeman, 2021; Potvin Kent et al., 2019; Smith et al., 2019, van der Bend et al., 2022c). For instance, research showed that social media influencers have a large impact on dietary behaviours in children, i.e., they increased their EDNP food intake when being exposed to less healthy food messages posted by influencers on Instagram and YouTube (Coates et al., 2019a, 2019b). Adolescents are particularly active on TikTok, enjoy short videos, and are susceptible to EDNP food marketing (Pechmann, Levine, Loughlin, & Leslie, 2005). It is crucial to find ways to increase recognition of the commercial intent of TikTok influencer videos and enhance critical attitudes among adolescents (Boerman, Willemsen, & Aa, 2017; Boyland & Tatlow-Golden, 2017).

This research aims to provide new insights into the effectiveness of a sponsorship disclosure in a TikTok influencer video with different levels of product placement, in terms of adolescents' recognition and understanding of and critical beliefs towards the commercial intent and brand outcomes.

2. Theoretical underpinnings

2.1. Dual processing mechanisms

An individual's processing of advertising messages has frequently been described or explained using dual processing models (Nairn & Fine, 2008), e.g. Petty and Cacioppo's Elaboration Likelihood Model (ELM) (Petty & Cacioppo, 1986). Dual processing models such as the ELM argue that central (explicit) or peripheral (implicit) processing routes are triggered when individuals are exposed to advertising messages (Nairn & Fine, 2008; Petty & Cacioppo, 1986). Implicit processing routes are triggered automatically and without cognitive effort, attention to or awareness of the message, and lead to more impulsive behaviours (Buijzen, Van Reijmersdal, & Owen, 2010). Thus, implicit attitudes may be developed without any intentional or explicit memory, and are rather related to increased perceptions of familiarity towards the product, which can result in liking or preferences for the product on the long term, referred to as the mere-exposure effect (Klinger & Greenwald, 1994; Yang & Roskos-Ewoldsen, 2007). Implicit attitudes have been shown to be a good predictor of behaviour (Fazio & Olson, 2003). On the contrary, explicit or systematic processing routes require cognitive effort, and lead to better recall and more deliberate decisions (Nairn & Fine, 2008). However, higher levels of explicit recall do not necessarily lead to liking or actual purchase or choice of a product (Gupta & Gould, 2007; Tulving, Schacter, & Stark, 1982).

Research suggests that the way persuasive commercial messages are being processed largely depends on the maturity or age of the individual processing them (Buijzen et al., 2010). Consumer and advertising skills undergo significant changes from early childhood to early adolescence (Buijzen et al., 2010). Previous research suggests that younger children under 12 years are more sensitive to less elaborate and hence more implicit processing mechanisms and have more cognitive difficulty processing persuasive messages. Adolescents from 12 to 14 years are generally more capable of activating knowledge on advertising tactics and critically reflecting on persuasive messages compared to younger children (John, 1999). Specifically, it is assumed that by the age of 16, adolescents' consumer- and advertiser-related skills and experience have reached adult-like levels (John, 1999). Yet, adolescents are still considered a vulnerable group when it comes to processing persuasive messages, as they are more impulsive, socially impressionable, and self-conscious, being in the midst of their identity development (van der Bend et al., 2022c; Pechmann et al., 2005). Since conforming to their peer group or to a certain sub culture is important in the adolescent phase, adolescents may be more susceptible to advertising messages that focus on brand-related social status, image, or physical appearance, and to messages embedded within their social networks (Buijzen et al., 2010; Pechmann et al., 2005). Thus, adolescents from 12 years old are at a critical cognitive developmental stage for processing social media

messages, and hence this age group is the main focus of the current study. Adolescents' understanding or recognition of persuasive intent, referred to as conceptual persuasion knowledge, can only be increased when they process a message more explicitly and with high cognitive elaboration (Buijzen et al., 2010). This may be a reason why an increasing number of marketers seem to exploit children's and adolescents' underdeveloped consumer skills and intentionally apply tactics designed to influence them through implicit processing mechanisms (Smith et al., 2019). Social media has led to a more complex commercial media environment for young people, where marketers use highly integrated tactics, causing advertising, information and entertainment to blend (Buijzen et al., 2010). The level of integration of food cues in advertising may impact how young people process these messages (Folkvord, Anschütz, Boyland, Kelly, & Buijzen, 2016). As such, social media marketing presents new challenges for young people's processing of advertising and may impair their ability to recognize or critically reflect on them (Buijzen et al., 2010).

2.2. Sponsorship disclosures

Sponsorship disclosures inform consumers about the commercial intent of messages, so that they can distinguish advertising from other (entertaining) content and understand the persuasive intent (Cain, 2011). Consequently, sponsorship disclosures may help increase consumers' critical beliefs towards the advertisement (Boerman, van Reijmersdal, & Neijens, 2012). Altogether, an individual's ability to activate strategies to defend against persuasive messages is referred to as persuasion knowledge. In the literature this term is used interchangeably with advertising literacy. The concept of persuasion knowledge has frequently been used to investigate young consumers' susceptibility to advertising effects (Naderer et al., 2018a; Wright, Friestad, & Boush, 2005). Having a higher conceptual persuasion knowledge is argued to be the first step towards being able to recognize and understand the persuasive intent of messages. Additionally, being able to develop critical beliefs, or hold a critical attitude towards the persuasive intent of a marketing message, is referred to as attitudinal persuasion knowledge, and this is suggested to be the second step in developing persuasion knowledge (Boerman, Reijmersdal, Rozendaal, & Dima, 2018). A higher attitudinal persuasion knowledge was found to negatively impact both brand desire and purchase intention in online social networking games (Rozendaal, Slot, van Reijmersdal, & Buijzen, 2013; Vanwesenbeeck, Ponnet, & Walrave, 2017). Moreover, persuasion knowledge can be separated into dispositional and situational persuasion knowledge, i.e., persuasion knowledge in general and in response to a specific instance of sponsored media content, respectively (Boerman et al., 2018).

It has been argued that it is particularly important to use sponsorship disclosures in social media posts involving an influencer or character, as posts by brand accounts would already automatically elicit a higher level of conceptual persuasion knowledge among consumers (Boerman et al., 2017). However, currently many sponsored influencer vlogs can be published without any disclosure of marketing intent, as in many countries, including the Netherlands, there are no legislative regulations on sponsorship disclosures that cover all social media influencer marketing content targeted to children and adolescents (Bridge, 2018; De, 2022; Meléndez-Illanes, González-Díaz, & Álvarez-Dardet, 2022).

Previous literature found conflicting evidence regarding sponsorship disclosure effects on adolescents' persuasion knowledge. For example, one study showed that advertising disclosure in sponsored vlogs increased adolescents' recognition of advertising and critical beliefs towards sponsored vlogs (De Jans et al., 2018). Another study showed that sponsorship disclosures in television marketing enhanced adolescents' understanding of persuasive intent while it did not mitigate persuasion or generate critical attitudes towards the brand (van Reijmersdal, Boerman, Buijzen, & Rozendaal, 2017). The study authors suggested that sponsorship disclosures may have a fundamentally different effect on adolescents compared to adults, as they would need more opportunities to process brand placements, e.g., through longer disclosure exposure (van Reijmersdal et al., 2017). Yet, evidence on the impact of sponsorship disclosures among adolescents is still limited, especially when used in the popular short-format type videos such as on TikTok, while this could significantly help protect adolescents against the harmful effects of unhealthy food marketing on these types of platforms.

In the current research we focus on a short-format TikTok marketing video involving nano-influencers who promote Doritos. Previous research showed that a 'parasocial connection', i.e. feeling close to a social media influencer without personally knowing them, may impact the effectiveness of an influencer social media marketing post (Sokolova & Kefi, 2020). Hence the peer influencers involved in this study are unknown to the adolescent viewers, since an already established parasocial connection may influence the effects of sponsorship disclosures differently among participants. Overall, we hypothesize the following.

H1. Influencer short-format social media videos with a sponsorship disclosure (vs no disclosure) lead to 1) more negative attitudes towards the brand promoted and 2) lower probability of choosing the product promoted (illustrated in Fig. 1).

We expect that these proposed direct associations may be (partly) explained by changes in adolescents' conceptual and attitudinal persuasion knowledge as a result of being exposed to the sponsorship disclosure. First of all, we hypothesize that conceptual persuasion knowledge levels increase as a result of exposure to sponsorship disclosure.

H2. Influencer short-format social media videos with a sponsorship disclosure (vs no disclosure) are more likely to lead to higher levels of conceptual persuasion knowledge in adolescents (illustrated in Fig. 1).

It has been suggested that young people who are able to understand the intention of persuasive messages are also more critical towards these types of messages (Boush, Friestad, & Rose, 1994; Wright et al., 2005). Thus, we hypothesize that conceptual persuasion knowledge positively impacts attitudinal persuasion knowledge.

H3. Higher levels of conceptual persuasion knowledge in adolescents will lead to an increase in their attitudinal persuasion knowledge (illustrated in Fig. 1).

Also, higher levels of persuasion knowledge suggests that more systematic processing is taking place, and this will impact the two outcome measures.

H4. When adolescents' attitudinal persuasion knowledge is increased, they 1) have more negative attitudes towards the brand promoted and 2) are less likely to choose the product promoted (illustrated in Fig. 1).

Yet, sponsorship disclosures will be most effective in increasing adolescents' conceptual and attitudinal persuasion knowledge when other cues in a persuasive commercial message might not help recognize it as advertising. Therefore, it is expected that the effectiveness of sponsorship disclosures may largely depend on the level of integration of the product or brand into a social media video.

2.3. Level of product-plot integration

Previous work suggests that marketing communications can be described along two different dimensions, i.e., 1) the degree to which the sponsor of the message or commercial intention of the message is disguised, and 2) the degree to which the persuasive message is secondary to the main message of communication (Nebenzahl & Jaffe, 1998). Product placement is a marketing communication strategy highly relevant to both television and digital media. It concerns the commercial, unobtrusive integration of a product into other media content (McCarty and Shrum, 2004). It can be a powerful strategy as it does not explicitly link the product in a scene to a brand or company, and the

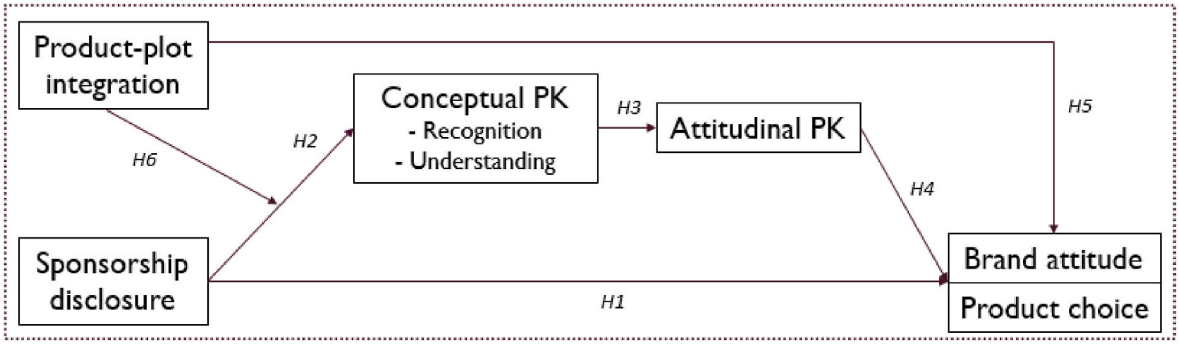


Fig. 1. Moderated mediation model proposing how social media food marketing messages with a low vs high product-plot integration differently impact the presence of a disclosure of sponsorship on adolescents' conceptual and hence attitudinal persuasion knowledge, consequently influencing brand attitude and product choice. *PK = Persuasion knowledge, H = Hypothesis. "Recognition" = Recognition of advertising, "Understanding" = Understanding of persuasive intent.*

persuasive message is often secondary to the main communication of the scene. Hence, product placement can often be characterized by its prominence or plot integration (La Ferle & Edwards, 2006; Lehu & É, 2009; McCarty and Shrum, 2004). Since product placement has a much more stealth nature than traditional advertising (e.g., commercial breaks during a movie), viewers may not immediately recognize the persuasion intent. Hence their persuasion knowledge, which would typically put them on guard in case of traditional advertising, is not enhanced (Wright et al., 2005).

There are many ways in which influencers can promote a product or brand in their social media videos, i.e., by applying either subtle or more obvious placements. The latter may involve showing, using, or verbally mentioning the product or brand. Specifically, previous studies used the degree of interaction between characters and food products, referred to as character-product interaction (CPI), as a measure for the level of product integration in a commercial message (Naderer et al., 2018a; Spielvogel, Matthes, Naderer, & Karsay, 2018). When brands are being handled or consumed by the main character in the video, also referred to as highly integrated or interactive brands placements, this leads to enhanced explicit memory and recognition of the brand (Kamleitner & Khair Jyote, 2013). Movement and higher placement prominence may explain these explicit effects (Abrams & Christ, 2003; Reijmersdal, 2009). Moreover, while CPI was found to increase brand recall, it also led to more positive brand attitudes than non-CPI or static placements (Kamleitner & Khair Jyote, 2013; Naderer et al., 2018a, 2018b; Spielvogel et al., 2018; Yang & Roskos-Ewoldsen, 2007).

Arguably, the effects of increased brand recall and recognition suggest that more explicit or systematic processing is taking place (Buijzen et al., 2010), which would expectedly lead to more critical processing. Yet, in the case of CPI, observed positive effects on brand attitude may be attributed to a parasocial connection. When adolescents feel a certain social connection with the influencer, and the influencer interacts with the product, viewers may develop positive attitudes towards the product due to an identification process (Naderer et al., 2018b). On the contrary, an influencer video may also trigger critical attitudes or beliefs towards the product, depending on the product's level of integration with the storyline of the video (Kamleitner & Khair Jyote, 2013; Russell, 2002). CPI automatically generates a certain level of connection between the product and the plot, but may still involve multiple ways of product use and hence degree of product-plot integration. The extent to which products are connected to the storyline of a message was found to play an important role in its effects (Russell, 2002). Specifically, messages that become too intrusive, i.e., with a high product-plot integration and thus an obvious focus on the product, are less likely to generate favourable effects on attitudes (Russell, 2002). Thus, we hypothesized the following.

H5. A higher product-plot integration in a short-format social media influencer video leads to 1) more negative attitudes towards the brand promoted and 2) lower probability of choosing the product promoted (illustrated in Fig. 1).

Subsequently, we hypothesize that a high product-plot integration already works as a heuristic cue that may increase conceptual persuasion knowledge in adolescents, and therefore a disclosure may have a stronger effect in social media videos with a low product-plot integration. Hence, it is expected that the level of product-plot integration will moderate the sponsorship disclosure's impact on persuasion knowledge, which consequently influences brand attitude and product choice.

H6. The level of product-plot integration in a short-format social media influencer video moderates the effect of the sponsorship disclosure (vs no disclosure) on conceptual persuasion knowledge: The sponsorship disclosure has a larger effect when the message has a lower product-plot integration (vs higher product-plot integration) (illustrated in Fig. 1).

3. Methodology

3.1. Study design & sample

This school-based experimental study involves a between-subjects 2 × 2 factorial design, with level of product-plot integration (static placement vs. high plot integration) and sponsorship disclosure (no disclosure vs. disclosure) as the main experimental independent variables. Data were collected from a convenience sample of 262 Dutch adolescents, between April 19, 2022 and May 25, 2022. The experiment was pre-registered in Open Science Framework (OSF) on June 24, 2022, after data collection, before viewing the data and conducting statistical analysis. High school teachers who were informal contacts of the researchers were contacted to recruit participants from the first four classes in high school (i.e., 12–17 years old), as these include all educational levels. Participants were randomly assigned to an experimental condition using block randomization, with roughly equal amounts of people being exposed to one of four experimental groups (see descriptions in section 3.3) using Qualtrics.

3.2. Study procedure

This study was approved by the Social Sciences Ethical Committee of University [X]. Participants were not made aware of the study purpose before participation, which was to determine the response to a TikTok influencer marketing video promoting the chips brand Doritos. TikTok is highly popular among the adolescent group, and there has also been much interest in this social media platform for influencer marketing purposes. Specifically, 70% of marketers identified TikTok as the main social media platform to focus their marketing efforts on, and increased their TikTok influencer marketing by 325% (Linqia, 2021).

In a cover story, participants were informed that they were going to

watch a 1-min TikTok video, and that the aim of the study was to investigate their opinions on the TikTok video and on social media in general. Before the study took place in high school classes, parents were also informed about the study through information letters, which gave them the opportunity to object to their child's participation in the study before the data collection took place (i.e., passive consent). On the day of data collection, two researchers visited the school classes and first gave a short explanation to the teachers. They were made aware of the study purpose but asked to not provide any details about this to the students beforehand. After giving a short introduction to a class, students within the class were allocated into groups of 5–8 participants, depending on the length of the lesson and number of students in the classroom. Each group was separated from the classroom where the main lesson was held and led to another empty classroom or study space with no background noise or distractions. Participants were asked to bring their own phones or devices they used for social media, and their own headphones if they had any. The two researchers present during data collection made sure to provide participants with spare headphones if they did not have their own, or chargers in case their phone battery was low. In case participants' phones did not work or there were not enough headphones for Apple or Android phones, researchers provided them with spare Android phones. Next, the researchers provided a QR code which directed participants to a Qualtrics survey on their phone. The first questions in the survey asked participants to give consent to participate in the study. At this point students were able to either accept or decline participation. Next, they watched a video of a compilation of three TikTok videos. They were told to only watch the video once, and then click through to the next page. The time participants spent on the page was measured using the timing function in Qualtrics. On the page after the video, the questionnaire included items on (in order) attention, brand attitude, brand familiarity and regular consumption, conceptual and attitudinal persuasion knowledge, disclosure recognition, brand recall, liking of the video and people in the video, regular watching frequency of short form videos, and demographics (age, gender, educational level). In the end participants saw a message saying the questionnaire had ended, and they were invited to pick one of six crisp brand flavours as a thank you gift (including the brand and flavour promoted in the TikTok video). This last choice question served as an implicit measure of product choice.

3.3. Stimulus material

The video shown to the participants was a compilation of three different TikTok videos that were merged into one video, which took in total 47 s. Each individual TikTok video took up about one third of the entire video. The compilation video was created by the researcher, who recorded their screen while scrolling through the three videos in TikTok. The first and third video were the same for all participants and showed a funny animal video and a video with people doing tricks, respectively. The second video was 18 s long and the main subject of study, although participants were not aware of this. It was chosen to add a video before and after the main video to make it look more realistic and less static (i.e., participants would get more the idea or feeling of scrolling through TikTok), and disguise the main study aim.

The video of main interest for the study - the second in the compilation - was created by 19-year-old communication students. For these students, this was an assignment part of their study curriculum for which they got study credits, and they were experienced in creating TikTok videos. They created the video with input from one of their siblings who was in the target group (girl, 14 years). In the video, two friends of the communication students, a 19-year-old boy and girl are playing the popular online game 'Among us' against each other on their laptop, while they are sitting next to each other on a couch. A blue bag of Doritos chips, flavour 'Cool American' is placed in between them. Both characters in the video fanatically try to win the game. In the beginning of the video, the text 'Tell me you're competitive without telling me you're competitive' is shown on the screen. This is a popular challenge which originally started on Twitter in 2019, in which people demonstrate or show something without literally explaining it in words or outright saying what they mean (Gabrielle, 2022). The TikTok video was shot partly from the front and the back of the couch, to be able to show viewers what game the characters were playing.

The video was manipulated in four different ways, and participants were randomly allocated to one of four experimental conditions (see illustrative screenshots of the video in Supplementary Material 1A-D).

1. Low product-plot integration: video in which the bag of Doritos crisps is placed in between the two characters and is clearly visible but not touched or consumed. Disclosure: No sponsorship disclosure is shown (Supplementary Material 1A).
2. High product-plot integration: video in which the bag of Doritos chips is initially placed in between the two characters and is then opened, consumed and thrown by the characters in the video as part of their competition and attempts to beat each other. As a result, the chips are also lying on the laptops of the characters and on the couch. Disclosure: no sponsorship disclosure is shown (Supplementary Material 1B).
3. Low product-plot integration: video in which the bag of Doritos chips is placed in between the two characters and is clearly visible but not touched or consumed. Disclosure: a sponsorship disclosure is shown (Supplementary Material 1C).
4. High product-plot integration: video in which the bag of Doritos chips is initially placed in between the two characters and is then opened, consumed and thrown by the characters in the video as part of their competition and attempts to beat each other. As a result, the chips are also lying on the laptops of the characters and on the couch. Disclosure: a sponsorship disclosure is shown (Supplementary Material 1D).

All videos played the same tune, which was picked from a list of tunes typically provided by the TikTok app. With regard to the product-plot integration, it was made sure that both videos (low, high) started and ended with the same shot (i.e., the bag of Doritos bag placed in between the characters), and the part in the middle showed a change in plot integration. With regard to the disclosure, previous work recommends certain requirements for disclosures to be effective with regard to generating persuasion knowledge (Rozendaal & van Reijmersdal, 2017). Namely, they should be visible in text on the screen, in a spot where it is easily noticeable, with clear description of whom is paid by what brand to make the content, and it should be longer than 6 s (e.g. 10 s) (Rozendaal & van Reijmersdal, 2017). A disclosure text in Dutch was added to the TikTok video after it was completed, saying "We are paid by Doritos to advertise in this video". It was prominently shown in the upper centre of the screen, for 10 s, and visible 4 s after the start of the video. In the first 4 s of the video, a voice said 'Tell me your competitive, without telling me your competitive', while the same words appeared on the lower centre of the screen. It was chosen to not show both texts (i.e., disclosure and challenge line) simultaneously as this would create too many stimuli at the same time and could result in participants not fully processing or seeing the disclosure.

Pre-testing of the video was carried out by showing it to three ($n = 3$) teenagers 12–14 years, who were asked to indicate whether they would like the video if they saw it on TikTok, and they confirmed this. After the video was finalized, the researcher posted the video on their own TikTok page and made it public for less than 2 h. Within this period, 6 likes were generated through different email addresses and contacts of the researcher, after which the video was made private again. Directly after this, the compilation of the three videos was created through screen recording by the researcher. This was done as quickly as possible, as the relatively low number of likes would make more sense when the video was just posted recently, and this would make it look less suspicious to the participants.

3.4. Measures

3.4.1. Brand attitude

To measure brand attitude, participants were told the brand Doritos was shown in the second TikTok video (in case participants were not able to answer the brand recall question). Next, they were asked for their opinion on the brand Doritos (1 = dislike very much, 6 = like very much). Additionally, they were asked to order three premium crisps brands (Doritos, Lays, Cheetos) in terms of liking. In this way, the relative liking of Doritos compared to other premium brands was also measured. So, Doritos got the highest score (3) when it was at the top of the list and the lowest score (1) when it was placed at the bottom of the list.

3.4.2. Product choice

After the 'official' questionnaire ended, there was one additional survey page on which participants were offered to pick a 'thank you'-gift as appreciation for their participation in the study. They were provided with images of in total six different chips bags of three premium brands (Doritos Cool American and Nacho Cheese, Lays Natural and Bugles Nacho Cheese, Cheetos Chipito and Nibb-its). A choice measure was considered to be relevant to the target group, as adolescents may still depend on their parents to buy things and hence measuring their intentions to purchase chips bags may not be realistic to them. Also, measuring intentions has a limited predictive value with regard to adolescents' actual choices, because they tend to act more spontaneously rather than through deliberative thought (Naderer et al., 2018b).

3.4.3. Conceptual persuasion knowledge

Two aspects of conceptual persuasion knowledge were measured with four items in total, and according to the same questions used by van Reijmersdal et al. (2017), who specifically developed the questions for 13–17 years old Dutch adolescents (van Reijmersdal et al., 2017). Instead of 7-point scales, 6-point Likert scales were used, as adolescents more frequently pick neutral options (e.g., neither agree nor disagree) (Raaijmakers, Hoof, Hart, Verbogt, & Vollebergh, 2000).

First of all, recognition of advertising was measured by asking participants to what extent (1 = strongly disagree, 6 = strongly agree) they believed the second TikTok video, in which the game 'Among us' was played, was advertisement.

Second, understanding of persuasive intent was measured by asking participants to indicate the extent to which they thought the TikTok video in which the game 'Among us' is played, was made to "make me like the brand", "sell Doritos", and "influence me" (1 = strongly disagree, 6 = strongly agree). The mean score of the three items was considered a sufficiently reliable measurement of the understanding of persuasive intent, as the value for Cronbach's Alpha was $\alpha = 0.70$.

For all analyses involving conceptual persuasion knowledge, effects were presented separately for recognition of advertising and understanding of persuasive intent, similar to previous authors (van Reijmersdal et al., 2017).

3.4.4. Attitudinal persuasion knowledge

Attitudinal persuasion knowledge was also measured in the same way as done by van Reijmersdal et al. (2017), with four items, i.e., by asking participants to what extent they agreed to statements about the TikTok video in which 'Among us' is played (1 = strongly disagree, 6 = strongly agree): "I think the video is "honest" (reversed), "trustworthy" (reversed), "convincing" (reversed), and "not credible". Again, Cronbach's alpha was determined to see whether the mean score of the four items could be used as a reliable measurement of attitudinal persuasion knowledge. Similar to the findings reported by van Reijmersdal et al. (van Reijmersdal et al., 2017), the Cronbach's alpha was relatively low when including the four items ($\alpha = 0.50$). However, after removing the item "not credible", the Cronbach's alpha value improved to $\alpha = 0.60$, which was considered sufficient (Nunnally, 1967). It is possible that the reverse scale for this particular item confused the participants.

3.4.5. Brand recall

Participants were told what the second video was about, i.e., a boy and girl playing the game 'Among us' on the computer. Next, participants were asked whether they recalled seeing any brands in this second TikTok video (Yes/No), and if yes, which brands (0 = did not mention Doritos, 1 = mentioned Doritos).

3.4.6. Control variables

In addition to the mediators and outcomes in the model, several control variables were measured to test randomization, or to check whether the effects of the disclosure and plot integration were not caused by other differences between the experimental groups. *Brand familiarity* was measured by asking participants whether they already knew the brand Doritos before seeing the video. *Regular brand consumption* was measured by asking participants how often they ate Doritos on average (1 = never, 2 = less than once a month, 3 = two to three times a month, 4 = weekly, 5 = Daily). *Liking of the video and characters* in the video was measured by asking participants to give their opinion (1 = dislike very much, 6 = like very much) on the following questions: "What do you think of the video in which the game 'Among us' is played?" and "What do you think of the girl and boy in the video?". *Watching frequency of short form social media videos* was measured by asking participants "How often do you watch short form videos on social media (think of TikTok videos, Instagram Reels, Facebook reels, or YouTube Shorts)?". Answer categories were: 1 = never, 2 = less than once a month, 3 = two to three times a month, 4 = weekly, 5 = Daily. Additionally, participants were asked about their *age*, *gender* and *educational level*. Finally, the *school locations* where the study was conducted were recorded as well.

3.4.7. Manipulation check

Participants' recognition or recall of the sponsorship disclosure was measured by asking those who were exposed to a disclosure whether they saw a textual disclosure in the second video, and if so which one, with the following options: the correct disclosure, two random disclosures that were related to the game 'Among us' specifically (i.e., not related to sponsorship or to the challenge 'Tell me without telling me'), and the option "I did not see any textual disclosure in the video".

3.4.8. Attention check

An attention check was conducted to determine whether participants paid attention to the experimental TikTok video. Participants were asked to indicate what topics were covered by the three TikTok videos they watched. They could pick three options from a list of six, of which three were referring to the correct three topics, two included imaginary topics and one "I don't know". Participants who did not select the topic of the experimental video (i.e., "A video in which a boy and girl play a game on the computer") as one of the three options, regardless of the other options they picked, were excluded from the analyses (n = 17).

3.5. Data analysis

Statistical analysis was conducted using SPSS Version 28.0.1.1. Participants who had not finished the survey up to and including the demographic questions (age, gender, educational level) were excluded ($n = 9$); participants who completed all questions except for product choice were included, and product choice was set as missing.

With regard to manipulation effectiveness, it was checked whether participants who were exposed to the disclosure (i.e., two experimental groups) had actually seen the (correct) disclosure. A similar approach was used by van Reijmersdal et al. (van Reijmersdal et al., 2017) and Boerman et al. (Boerman et al., 2012).

Furthermore, before testing any of the main hypotheses, randomization was checked, by performing Chi Square analyses or analyses of

variance to see whether the four experimental groups differed significantly with respect to gender, age, educational level, school location, brand familiarity, regular brand consumption, relative brand attitude, liking of the video and characters, attention to the video, and watching frequency of short form videos on social media. Tests were considered significant when $p < .05$. Based on this, it was decided which covariates were added into the models that were used to test the hypotheses.

To assess how the sponsorship disclosure impacts brand attitude (H1), its main effect was determined in a two-way ANOVA, in which sponsorship disclosure and level of product-plot integration were entered as independent variables and brand attitude as outcome. To determine the disclosure's impact on product choice, logistic regression was performed. Furthermore, the disclosure's effect on conceptual persuasion knowledge (i.e., recognition of advertising and understanding of persuasive intent) was assessed using a two-way MANOVA (H2).

To test mediation effects by conceptual and attitudinal persuasion knowledge in serial (H3 and H4), Model 80 of the PROCESS version 4.1 in SPSS (Hayes, 2022) was applied. Thus, it was determined how a sponsorship disclosure (vs no disclosure) shown in an influencer short-format social media video impacts adolescents' conceptual persuasion knowledge, and consequently attitudinal persuasion knowledge, and to what extent attitudinal persuasion subsequently affects brand attitudes and product choice.

The main effect of level of product-plot integration on brand attitude (H5) was assessed by means of a two-way ANOVA, and its impact on product choice was tested by means of a logistic regression.

To test H6, i.e., the moderating effect of level of product-plot integration on the association between sponsorship disclosure and conceptual persuasion knowledge (i.e., recognition of advertising and understanding of persuasive intent), a two-way MANOVA was conducted, with the interaction term as independent variable.

Lastly, to examine whether the effects of a sponsorship disclosure on conceptual, attitudinal persuasion knowledge, and consequently brand attitudes and product choice differ for levels of product-plot integration, a moderated multiple mediation analysis was conducted using PROCESS Model 85 (Hayes, 2022). Since this moderated mediation model does not allow for parallel mediators, as the first mediator recognition of advertising and subsequently understanding of persuasive intent were put into the model in separate analysis, for the two outcome variables.

4. Results

4.1. Sample characteristics and randomization check

A total of 245 Dutch adolescents with a mean age of 13.6 years (SD = 1.42) participated in the study (Table 1). None of the (parents of the) adolescents declined participation. The sample included 128 girls, 111 boys, and 6 indicated 'other' gender. Most participants followed a university preparatory educational level (59%). Furthermore, a large proportion of the sample was from high schools located in Amsterdam (43%) and Wageningen (43%).

The four experimental groups did not differ significantly ($p < .05$) with respect to age, $F_{(3,\ 241)} = 0.731$, $p = .534$, gender, $\chi^2_{(6)} = 4.982$, $p = .546$, educational level, $\chi^2_{(6)} = 2.358$, $p = .884$, brand consumption frequency, $F_{(3,\ 241)} = 1.724$, $p = .163$, watching frequency short-format videos, $F_{(3,\ 241)} = 1.415$, $p = .239$, school location, $\chi^2_{(12)} = 4.182$, $p = .980$, liking of the video, $F_{(3,\ 241)} = 1.085$, $p = .356$, liking of the characters, $F_{(3,\ 241)} = 1.599$, $p = .190$, relative brand liking, $F_{(3,\ 241)} = 0.158$, $p = .925$, brand familiarity, $\chi^2_{(3)} = 6.410$, $p = .093$. Therefore, it was concluded that the experimental groups were successfully randomized, and none of the control variables were added as covariates to the analyses.

4.2. Manipulation check

Of all 120 participants exposed to the sponsorship disclosure (49%), 46% ($n = 55$) had recognized the correct disclosure, 18% ($n = 21$) chose one of the two alternative, incorrect disclosures, and 37% (n = 44) said to have not seen any textual disclosure.

Table 1
Characteristics of the total sample and for each experimental condition (i.e., level of product-plot integration, sponsorship disclosure).

N	Total sample	Low product-plot integration		High product-plot integration	
		No disclosure	Disclosure	No disclosure	Disclosure
	245	62	61	63	59
Age in years, M (SD)	13.6 (1.42)	13.7 (1.37)	13.7 (1.43)	13.7 (1.45)	13.4 (1.41)
Gender, n (%)					
Male	128 (52)	30 (48)	29 (48)	36 (57)	33 (56)
Female	111 (45)	31 (50)	29 (48)	25 (40)	26 (44)
Other	6 (2)	1 (2)	3 (5)	2 (3)	0 (0)
Educational level, n (%)					
Preparatory vocational	19 (8)	7 (11)	3 (5)	4 (6)	5 (8)
Senior general	81 (33)	18 (29)	22 (36)	21 (33)	20 (34)
University preparatory	145 (59)	37 (60)	36 (59)	38 (60)	34 (58)

Table 2
Means of recognition of advertising, understanding of persuasive intent, attitudinal persuasion knowledge, brand attitude, and frequencies of product choice and brand recall for the disclosure conditions.

N	Total sample	No disclosure	Disclosure
	245	125	120
Recognition of advertising, M (SD)	3.81 (1.25)	3.65 (1.15)	3.98 (1.33)***
Understanding of persuasive intent, M (SD)	3.95 (1.06)	3.72 (1.09)	4.2 (0.97)*
Attitudinal persuasion knowledge, M (SD)	4.0 (0.91)	4.11 (0.91)	3.90 (.89)
Brand attitude, M (SD)	4.88 (.80)	4.90 (0.79)	4.88 (0.83)
Product choice, n (%)[a]	50 (21)	25 (21)	25 (21)
Brand recall, n (%)	125 (51)	35 (28)	90 (75)***

[a] $n = 7$ missing values for this variable, percentages are calculated based on all non-missing cases.
[b] Significantly different compared to no disclosure: *$p < .05$, **$p < .01$, ***$p < .001$.

4.3. The effects of sponsorship disclosure

Direct effects of sponsorship disclosure on the measures of persuasion knowledge and brand outcomes are presented in Table 2. With regard to H1, it was found that the sponsorship disclosure did not significantly affect **brand attitude** ($F_{(241)} = 0.02$, $p = .894$). Also, the presence of the disclosure did not significantly decrease probability of **product choice** ($b = 0.04$, $OR = 1.04$, $p = .894$), compared to no disclosure (−2 Log Likelihood = 244.68, Nagelkerke $R^2 = 0$, χ^2 (1) = 0.02). Thus, H1 was rejected. Logistic regression showed a significant and positive effect of disclosure on brand recall ($b = 2.04$, $OR = 7.71$, $p = < .001$), compared to no disclosure (−2 Log Likelihood = 283.2, Nagelkerke $R^2 = 0.274$, χ^2 (1) = 56.34).

For H2, the analyses demonstrated that sponsorship disclosure had a significant, positive effect on the two measures of conceptual persuasion knowledge (Wilk's Lambda = 0.95, $F_{(240)} = 6.65$, $p = .002$, eta^2 = 0.052), as it increased both **recognition of advertising** ($F_{(241)} = 4.48$, p

= .035) and **understanding of persuasive intent** ($F_{(241)} = 13.28$, $p = <.001$). This is in line with our hypothesis.

PROCESS analyses conducted to assess the mediation effects proposed in Fig. 1 showed no *indirect* effects on **brand attitude** via recognition of advertising, understanding of persuasive intent, or attitudinal persuasion knowledge, or via either recognition of advertising or understanding of persuasive intent in serial with attitudinal persuasion knowledge (Table 3). Overall, the disclosure did not have a significant *total* effect on **brand attitude** in the total sample. Thus, mediation effects of disclosure on **brand attitude** were not in line with our hypotheses H3 and H4.

Additional mediation analyses regarding H3 and H4 showed a significant, positive *indirect* effect of disclosure on **product choice**, via understanding of persuasive intent (effect = 0.20, boot se = 0.11, CI 0.01; 0.45), which is the opposite effect of what was hypothesized. Furthermore, *indirect* effects of disclosure on **product choice** via either recognition of advertising or understanding of persuasive intent in serial with attitudinal persuasion knowledge, were not significant (p > .05). Thus, the proposed mediation effect of conceptual persuasion knowledge and subsequently attitudinal persuasion knowledge was not supported.

4.4. Level of product-plot integration

Table 4 shows the means of the measures of persuasion knowledge and brand outcome variables, for the two conditions of product-plot integration. With regard to H5, level of product-plot integration was found to have no significant effect on **brand attitude** ($F_{(241)} = 0.52$, $p = .473$) and **product choice** ($b = -0.059$, $OR = 0.943$, $p = .854$). Furthermore, no significant effect was found of product-plot integration on brand recall (p < .05).

Also, no significant effect was observed of the level of product-plot integration on recognition of advertising and understanding of persuasive intent: Wilk's Lambda = 0.99, $F_{(240)} = 1.02$, $p = .364$, $eta^2 = 0.008$. Regarding H6, the interaction effect of the disclosure and level of product-plot integration on **recognition of advertising** and **understanding of persuasive intent** was not significant (Wilk's Lambda = 0.98, $F_{(240)} = 2.90$, $p = .057$, $eta^2 = 0.024$). Thus, the disclosure did not have a significantly larger effect on the two conceptual persuasion knowledge measures when used in a video with a low product-plot integration, compared to the video with a high product-plot integration. Therefore, H6 was not supported. Fig. 2 shows all direct effects of disclosure, moderated by level of product-plot integration, on all mediators and outcomes.

The mediation analysis on the effects showed no significant indirect effect of the interaction of disclosure and level of product-plot integration via the measures of persuasion knowledge on **brand attitude** and **product choice** (Table 5).

Table 4
Means of recognition of advertising, understanding of persuasive intent, attitudinal persuasion knowledge, brand attitude, and frequencies of product choice for the conditions of product-plot integration.

N	Total sample	Low product-plot integration	High product-plot integration (total sample)
	245	123	122
Recognition of advertising, M (SD)	3.81 (1.25)	3.83 (1.31)	3.80 (1.20)
Understanding of persuasive intent, M (SD)	3.95 (1.06)	4.04 (1.03)	3.87 (1.08)
Attitudinal persuasion knowledge, M (SD)	4.0 (0.91)	3.95 (0.89)	4.06 (0.92)
Brand attitude, M (SD)	4.88 (0.80)	4.92 (0.75)	4.84 (0.85)
Product choice, n (%)[a]	50 (21)	26 (21)	24 (21)
Brand recall, n (%)	125 (51)	58 (47)	67 (55)

*Significantly different compared to no disclosure: *$p < .05$, **$p < .01$, ***p < .001.

[a] $n = 7$ missing values for this variable, percentages are calculated based on all non-missing cases.

5. Discussion

5.1. Main findings of this study

The current study investigated the effect of the presence of a prominent sponsorship disclosure in a TikTok influencer marketing video with a low versus high product-plot integration of the chips brand Doritos into the storyline on adolescents' persuasion knowledge, and consequently their attitude towards the promoted product, and product choice. Overall, the analyses demonstrated that the disclosure had significant direct and indirect effects on some of the mediators and outcomes measured in this study, while the level of product-plot integration did not have any significant direct or indirect effect. No moderation effect of product-plot integration on the relationship between sponsorship disclosure and the mediators and outcomes was observed.

5.1.1. Sponsorship disclosure

The current study investigated a textual, prominent sponsorship disclosure, which was placed in the upper part of the TikTok video for a duration of 9 s. This disclosure was developed in line with sponsorship disclosure criteria that were established previously by other authors (Rozendaal & van Reijmersdal, 2017), who also provided specific suggestions for the wording. This study's findings showed that, with regard to direct effects, the disclosure had a significant and positive impact on conceptual persuasion knowledge (i.e., both recognition of advertising and understanding persuasive intent), and thus made the advertising and persuasive intent more transparent to the adolescents. These effects are in line with several previous studies on television or online marketing that showed a positive impact of disclosure on conceptual

Table 3
Mediation effects of disclosure on brand attitude and product choice in the total sample, via either recognition of advertising or understanding of persuasive intent, and attitudinal persuasion knowledge.

Type of effect	Independent variable	Mediator	Outcome variable	*b*	SE	BCA95%CI
Direct	Disclosure	-	Brand attitude	−0.02	0.11	−0.23; 0.19
			Product choice	−0.07	0.33	−0.72; 0.58
Indirect	Disclosure	Recognition of advertising > attitudinal persuasion knowledge	Brand attitude	0.002	0.003	−0.001; 0.01
			Product choice	−0.005	0.008	−0.02; 0.006
Indirect	Disclosure	Understanding of persuasive intent > attitudinal persuasion knowledge	Brand attitude	−0.001	0.003	−0.01; 0.03
			Product choice	0.004	0.01	−0.01; 0.03
Total	Disclosure		Brand attitude	−0.01	0.10	−0.22; 0.19
			Product choice	–	–	-

SE = standard error; BCA95%CI = bias corrected and accelerated 95% confidence interval.

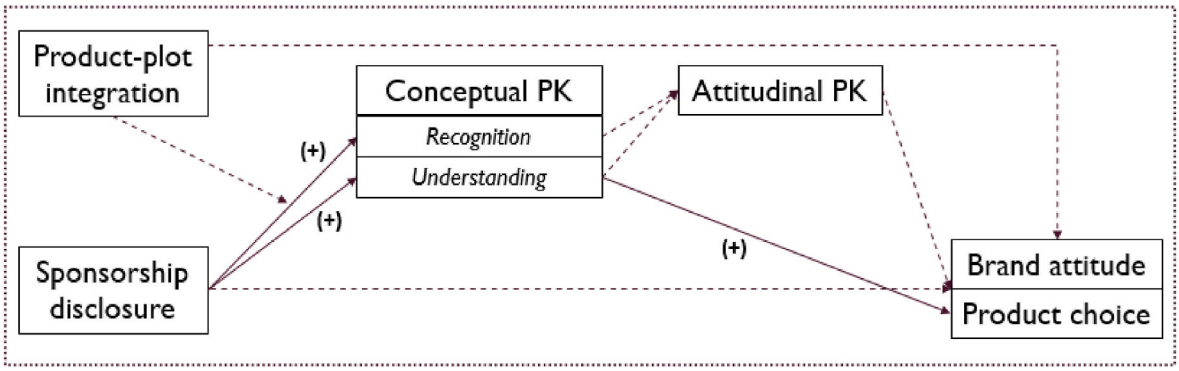


Fig. 2. Overall moderated mediation model showing all direct, significant effects ($p < .05$) of level of product-plot integration and presence of a disclosure of sponsorship on adolescents' conceptual and attitudinal persuasion knowledge, and on brand attitude and product choice. *PK = Persuasion knowledge, H = Hypothesis. "Recognition" = Recognition of advertising, "Understanding" = Understanding of persuasive intent. (−) Significant negative effect, (+) Significant positive effect. Dotted lines indicate no significant effect.*

Table 5
Moderated mediation effects of disclosure on brand attitude and product choice, via either recognition of advertising or understanding of persuasive intent, and attitudinal persuasion knowledge, by level of product-plot integration.

Type of effect	Independent variable	Mediator	Outcome variable	*Mod-med index*	SE	BCA95%%CI
Direct	Discl * integr	-	Brand attitude	0.01[a]	0.21	−0.40; 0.42
			Product choice	0.24	0.65	−1.04; 1.53
Indirect	Discl * integr	Recognition of advertising > attitudinal persuasion knowledge	Brand attitude	0.002	0.003	−0.003; 0.01
			Product choice	−0.004	0.009	−0.03; 0.008
Indirect	Discl * integr	Understanding of persuasive intent > attitudinal persuasion knowledge	Brand attitude	<0.001	0.002	−0.004; 0.003
			Product choice	<0.001	0.005	−0.009; 0.01

Discl. = Disclosure; Integr. = Level of product-plot integration; Modmed index = moderated mediation index, which is an unstandardized regression coefficient of the moderated mediation effect; SE = standard error; BCA95%CI = bias corrected and accelerated 95% confidence interval.
[a] Unstandardized b-coefficient.

persuasion knowledge, including recognition of advertising and/or understanding of persuasive intent (Boerman, 2020; Boerman et al., 2012, 2017; De Jans et al., 2018; De Jans & Hudders, 2020; De Veirman & Hudders, 2020; Matthes and Naderer, 2016, van Reijmersdal et al., 2017). Some of these studies also found effects of the disclosure on attitudinal persuasion knowledge and/or brand attitudes. However, in the current study neither the disclosure nor a higher conceptual persuasion knowledge led to an increase in attitudinal persuasion knowledge and/or decrease in brand attitude. This contradicts the notion that an increase in conceptual persuasion knowledge as a result of exposure to sponsorship disclosure helps adolescents to cope accordingly with an advertisement through attitudinal persuasion knowledge (Rozendaal, Lapierre, van Reijmersdal, & Buijzen, 2011), and hence mitigate brand attitude effects that lead to behavioural changes (Boerman et al., 2012). However, the findings showed that attitudinal persuasion knowledge and brand attitude were already relatively high across all conditions, and around 1/5 of the sample or more chose the product promoted, which is relatively high as there were six products to choose from. The fact that attitudinal persuasion knowledge was already high regardless of the manipulation may be explained by the experimental setting; participants were likely to be more on guard than when they use TikTok in a real-world setting. Regarding the high brand attitude, this may be attributed to the fact that brands play a particularly relevant role in adolescents' lives, as they strive to develop their own identity and are sensitive to achieving social status (Chaplin & John, 2005; Pechmann et al., 2005). Adolescents may already have strong brand attitudes towards Doritos, as it could have an important social meaning and therefore help adolescents 'fit in' into their peer group. Hence, adolescents could have had relatively positive implicit associations with Doritos, and rather followed gut reactions that guided them to a certain product choice (Gawronski & Bodenhausen, 2006; Matthes & Naderer, 2015). Hence, brand attitude and product choice may not have changed significantly by exposure to just one TikTok video.

Several studies have found conflicting results when it comes to the disclosure's ability to mitigate persuasion (Boerman et al., 2017; Matthes & Naderer, 2016; van Reijmersdal et al., 2017; van Reijmersdal & van Dam, 2020). Like this study, van Reijmersdal et al. (2017) concluded that a disclosure on television only had limited effects on persuasion knowledge, and did not mitigate persuasion (van Reijmersdal et al., 2017). Another study conducted among individuals in early (12–14 years) and mid (15–16 years) adolescence observed that the adolescent phase participants are in is crucial with respect to the effectiveness of a disclosure (van Reijmersdal & van Dam, 2020). Namely, the authors suggest that in early adolescence more explicit information on advertising and intention of a post is needed for the disclosure to increase their conceptual persuasion knowledge than in mid adolescence, as older adolescents will have an increased understanding of influencers' persuasive intent just by showing a disclosure of advertising (van Reijmersdal & van Dam, 2020). This supports other literature, which suggests adolescents' consumer and advertising skills to be underdeveloped until the age of 16 (Buijzen et al., 2010), and that younger adolescents have less well-developed persuasion knowledge (Verhellen, Oates, De Pelsmacker, & Dens, 2014). Those findings may partially explain the non-significant findings for attitudinal persuasion knowledge in the current sample. Most participants in this study (73%) were in early adolescence (12–14 years), which may explain the fact that no mitigation of persuasion took place, i.e. no significant changes in attitudinal persuasion knowledge, brand attitude and product choice were observed.

Regarding indirect effects of the disclosure on any of the outcomes, this study observed one remarkable effect in the total sample. Namely, it was observed that the disclosure led to a significantly higher chance of choosing the Doritos product, indirectly via a higher understanding of persuasive intent, which was the opposite of what was expected.

However, it should be noted that this effect was relatively small (effect = 0.20, boot se = 0.11, CI 0.01; 0.45). Interestingly, previous research suggests that influencers' clear communication about the advertising intent of their posts by using a sponsorship disclosure increases credibility of the influencer (Sesar, Martinčević, & Boguszewicz-Kreft, 2022; Weismueller, Harrigan, Wang, & Soutar, 2020). Thus, followers may perceive influencers as more credible when they display advertising status, and this impacts their purchase intention positively (Sesar et al., 2022; Weismueller et al., 2020). Thus, it is worthwhile to consider the conflicting effects that disclosures may have when used in the case of influencer posts. For example, disclosures may lead to higher persuasion knowledge and more negative product outcomes, or they may be a sign of transparency and trustworthiness and lead to more favourable product outcomes. Similarly, other work suggests that in case of stealthy product placements, such as sponsored influencer posts, being transparent about the posts' advertising nature, regardless of the commercial relationship, may result in less feelings of deception and consequently less negative consumer responses (De Veirman & Hudders, 2020; Evans, Wojdynski, & Grubbs Hoy, 2019).

5.1.2. Level of product-plot integration

The non-significant effects found for level of product-plot integration were not in line with previous studies on the effect of product-plot integration (Russell, 2002; Yang & Roskos-Ewoldsen, 2007). Based on literature, it was expected that, if adolescents do not know certain influencers, then they have not developed any parasocial connections towards these influencers yet, and hence a high product-plot integration would lead to higher levels of persuasion knowledge and more negative attitudes towards the product and lower chance of product choice, compared to a low product-plot integration. It is possible that the plot integration manipulation was not successful, as none of the mediators and outcomes were significantly impacted.

Similar to the findings by Hoek, Rozendaal, van Schie, van Reijmersdal, and Buijzen (2020), it could have been that the relatively equal product prominence in the low versus high product-plot condition led to non-significant differences in effects between the two conditions (Hoek et al., 2020). Brand recall was found to be already relatively high in the low product-plot condition (i.e., 47%) and differed non-significantly from the high product-plot condition (i.e., 55%). It may be that exposure to a package of the brand Doritos in both conditions, in an equally prominent manner, led to relatively similar participant responses. Indeed, it was made sure that in both conditions the Doritos package and logo was equally well visible or close to the camera and shown in the video in equal duration, to make sure that one of these factors would not affect the actual manipulation to be studied, i.e. the degree to which the product was integrated into the plot.

Furthermore, the large amount of information that adolescents were exposed to when watching the TikTok video could have affected the results. Specifically, in case of cognitive overload, a message will most likely not be processed at all, or merely automatically (Gao & Lang, 2009). Specifically, it is argued that younger adolescents are unable to 'stop and think' when they are overloaded with other elements related to a persuasive message, which limits their cognitive capacities (Rozendaal et al., 2011). According to the 'stop and think theory', for adolescents to use their persuasion knowledge as a critical defence, they need to first be able to stop to recognize and understand that the video is an advertisement. Secondly, they need to think about the commercial content which helps create potential critical beliefs to defend themselves (Rozendaal et al., 2011). The adolescents in this study may have been exposed to an overload of elements whilst watching the TikTok video. First, adolescents may have looked at the caption of the video, and the amount of likes and comments on the video which could have limited their cognitive capacity to think about the commercial content. Additionally, a woman's voice (i.e. default voice in TikTok) said *"Tell me you're competitive without telling me you're competitive"*, while the text was on screen. Previous research showed that memory is greater when a stimulus is spoken, as auditory information is more meaningful and thus processed more elaborately than visual information (Russell, 2002). Participants may have been distracted by the voice and may not have paid sufficient attention to the disclosure. Additionally, the effect of the experimental video may have been attenuated as participants could have been distracted by the two other TikTok videos shown before and after the experimental video. At the same time, this is a true representation of TikTok, where adolescents are exposed to a blend of entertaining content and information within a very short period of time.

5.2. Strengths & limitations

Several studies have been published on influencer marketing and its effects in both television and digital media. To our knowledge, this is the first study to investigate the impact of TikTok influencer marketing on persuasion knowledge and brand outcomes in adolescents. The research setting of this experiment aimed to represent a real world experience of using TikTok as optimally as possible by showing adolescents a compilation of three TikTok videos, presented in a screen recording video. Yet, this setting did not allow adolescents to scroll through the three TikTok videos themselves, as these were presented via a video in a Qualtrics questionnaire rather than a TikTok app. As a consequence, adolescents may have paid more of attention to the video than in a real-life situation, as they had to watch the entire video without being able to scroll to the next TikTok video. This may have led to more critical evaluations of the video. Alternatively, the adolescent participants may have been distracted by the unconventional experimental setting, i.e., the gathering of small groups of students who were guided to a separate class room, and who were allowed to use their own phone for the study. Also, participants may have distracted others while completing the survey. Yet, the study took place in a social setting, i.e., among other peers, and this may also reflect the setting in which adolescents sometimes scroll through their social media.

Another drawback of this study is related to the manipulation of the independent variables in this study. First of all, the effectiveness of manipulating product-plot integration was not pre-tested and also no manipulation check was conducted afterwards. Moreover, a disclosure manipulation check was only performed for the group that was exposed to the disclosure. While it was observed that only a sub-group (46%) of the adolescents who were exposed remembered the correct disclosure, analyses were still conducted on the full sample. It was chosen to do no additional analyses on the sub-group, as it would have been a significantly smaller group compared to the control group, and comparison of this sub-group to the control group for which this manipulation check was not performed could result in bias. Namely, it could lead to selection of adolescents who were relatively more alert and hence potentially had a higher persuasion knowledge. Yet, one could question whether a manipulation check can fully reflect the effectiveness of a manipulation, as priming research shows that cues, in this case sponsorship disclosures or level of product-plot integration, are not always processed consciously but can drive individuals' decisions regardless (Custers & Aarts, 2005).

Besides that, the way the survey was presented may have influenced the participants' responses. Namely, after the brand recall question participants were told that Doritos was shown in the second video. Also, all follow-up questions were about the second video in particular. This may have influenced advertising recognition or other persuasion knowledge measures.

This study also has some limitations regarding the adolescent sample and its representativeness of the adolescent group as a whole. Firstly, the sample was not equally distributed across educational levels, since the minority of adolescents (8%) followed a preparatory vocational secondary educational level. Persuasion knowledge may be affected by educational level, i.e., higher education was shown to be associated with more critical thinking (Boerman et al., 2018). Hence, a more representative group of adolescents would have allowed for better comparisons

across educational levels. Furthermore, most adolescents included in this study were 14 years or younger. Age may have a large impact on adolescents' processing skills and persuasion knowledge (Boerman et al., 2018; Buijzen et al., 2010), and hence future studies could consider including both a balance of younger and older adolescents.

Another drawback of this study is related to the brand that was chosen for the experimental video. As mentioned earlier, the adolescents in this study generally had a positive brand attitude towards Doritos. Relative brand attitude towards Doritos (i.e., compared to Cheetos and Lays) was also high, as only 8.6% of adolescents expressed Doritos to be their least liked brand. Thus, Doritos was a popular brand among adolescents, and they may have had pre-existing preferences or associations with Doritos, which could have been deeply ingrained. This could explain the insignificant direct and indirect effects found on brand attitude and product choice. Future studies could account for existing brand attitudes by either measuring these with ample time before exposing participants to the experimental video, or by measuring brand attitudes towards different food brands in a different sample that is large enough. Ideally, one would select a food brand with rather neutral attitudes. Future studies could also investigate the effects of unknown or fictitious brands, or even unbranded, generic products, to account for the relation between brand familiarity and product choice, and brand attitude. Additionally, product choice may have been affected by taste preference (i.e., Doritos Nacho Cheese and Lay's Naturel are incomparable flavours). Therefore, future studies could also consider taste preferences as a control variable. Moreover, it can be of interest for future studies to include other (savoury) snacks in the choice measure. By adding in different snacks, adolescents' general impulsivity towards crisps can be measured. Despite all above considerations, by choosing a rather implicit measure of product choice, i.e., presenting Doritos as an appreciation gift after participation in the study, this fits well with the increased impulsivity that characterizes the adolescent phase (Pechmann et al., 2005).

5.3. Conclusions and implications

It can be concluded that the presence of a textual sponsorship disclosure increases transparency of the commercial and persuasive intent of a TikTok influencer marketing video, as adolescents were better able to recognize that the video was an advertisement, and understood better that the video had a persuasive nature. These effects were found regardless of a low or high integration of the product into the storyline. This suggests that social media food marketing policies requiring sponsorship disclosure could be an effective first step towards creating more transparent TikTok influencer food marketing content. Specifically, it is recommended that brands and influencers aiming to advertise on TikTok apply a clearly visible textual disclosure in a similar, prominent manner as applied in the current study. This is particularly essential as current TikTok influencer practices do not seem to provide enough transparency on commercial intent (Morrison, 2022). Influencer content should integrate sponsorship disclosures, since marketing communications coming from brands already sufficiently signal their persuasive intent (Boerman et al., 2017). Particularly nano-influencers with lower numbers of followers currently do not fall under the Dutch influencer marketing regulations, and should therefore be required to also use disclosures. Yet, the current study found that increased transparency by the disclosure did not lead to more critical beliefs, attitudes, or altered product choice in adolescents. This suggests that, as posed by other theories, adolescents have more difficulty processing brand placements, as a result of their limited executive functioning skills (i.e., limited critical reflection or 'stop and think' response), and their identity formation and related susceptibility to social image, social status and brand symbolism (John, 1999; Rozendaal et al., 2011, van Reijmersdal et al., 2017). Yet, it is essential that research is continued on other, more optimal disclosure formats (e.g., wording, modality, duration, location) that increase transparency and subsequently generate behavioural impact. Moreover, investigating factors such as source (i.e. type of influencer) or type of product (i.e., generic, fictitious brand), along with disclosure effects may provide additional insights into the effectiveness of disclosures on TikTok. To our knowledge, the current research provides the first available evidence to inform future policies on the effects of protective sponsorship disclosure in popular short-format social media marketing videos targeted to adolescents. However, the current research also suggests that such policies may not be enough to mitigate persuasion, and perhaps the alternative policy focus should aim to protect adolescents against exposure to unhealthy food cues by restricting the promotion of unhealthy foods and beverages through brands and influencers on TikTok.

CRediT author statement

Daphne LM van der Bend: Conceptualization, Methodology, Formal analysis, Investigation, Writing – Original draft, Visualization, Project administration. **Nerine Gijsman:** Methodology, Formal analysis, Investigation, Writing – Original draft, Writing – Revies & Editing. **Tamara Bucher:** Writing – Review & Editing. **Vanessa A Shrewsbury:** Writing – Review & Editing. **Hans van Trijp:** Writing – Review & Editing. **Ellen van Kleef:** Conceptualization, Writing – Review & Editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chb.2023.107723.

References

Abrams, R. A., & Christ, S. E. (2003). Motion onset captures attention. *Psychological Science, 14*(5), 427–432.
Boerman, S. C. (2020). The effects of the standardized instagram disclosure for micro- and meso-influencers. *Computers in Human Behavior, 103*, 199–207.
Boerman, S., Reijmersdal, E., Rozendaal, E., & Dima, A. (2018). Development of the persuasion knowledge scales of sponsored content (PKS-SC). *International Journal of Advertising, 37*, 1–27.
Boerman, S. C., van Reijmersdal, E. A., & Neijens, P. C. (2012). Sponsorship disclosure: Effects of duration on persuasion knowledge and brand responses. *Journal of Communication, 62*(6), 1047–1064.
Boerman, S. C., Willemsen, L., & Aa, E. P. V. D. (2017). @ this post is sponsored : Effects of sponsorship disclosure on persuasion knowledge and electronic word of mouth in the context of Facebook. *Journal of Interactive Marketing, 38*, 82–92.
Boush, D. M., Friestad, M., & Rose, G. M. (1994). Adolescent skepticism toward TV advertising and knowledge of advertiser tactics. *Journal of Consumer Research, 21*(1), 165–175.
Boyland, E., & Tatlow-Golden, M. (2017). Exposure, power and impact of food marketing on children: Evidence supports strong restrictions. *European Journal of Risk Regulation, 8*(2), 224–236.
Bridge, M. (2018). *PewDiePie and other vloggers fail to disclose adverts*. The Times.
Buijzen, M., Van Reijmersdal, E. A., & Owen, L. H. (2010). Introducing the PCMC model: An investigative Framework for young people's processing of commercialized media content. *Communication Theory, 20*(4), 427–450.
Cain, R. M. (2011). Embedded advertising on television: Disclosure, deception, and free speech rights. *Journal of Public Policy and Marketing, 30*, 226–238.
CBS (2020). Jeugdmonitor. Internet jongeren, toegang. (2012-2019). *Gebruik en faciliteiten*.
Chaplin, L., & John, D. (2005). The development of self-brand connections in children and adolescents. *Journal of Consumer Research, 32*, 119–129.
Coates, A. E., Hardman, C. A., Halford, J. C. G., Christiansen, P., & Boyland, E. J. (2019a). Social media influencer marketing and children's food intake: A randomized trial. *Pediatrics, 143*(4).

Coates, A. E., Hardman, C. A., Halford, J. C. G., Christiansen, P., & Boyland, E. J. (2019b). The effect of influencer marketing of food and a "protective" advertising disclosure on children's food intake. *Pediatr Obes, 14*(10), Article e12540.
Custers, R., & Aarts, H. (2005). Positive affect as implicit motivator: On the nonconscious operation of behavioral goals. *Journal of Personality and Social Psychology, 89*(2), 129–142.
DDMA. (2022). *De Mediawet*. Available from: https://influencerregels.com/mediawet/.
De Jans, S., Cauberghe, V., & Hudders, L. (2018). How an advertising disclosure alerts young adolescents to sponsored vlogs: The moderating role of a peer-based advertising literacy intervention through an informational vlog. *Journal of Advertising, 47*(4), 309–325.
De Jans, S., & Hudders, L. (2020). Disclosure of vlog advertising targeted to children. *Journal of Interactive Marketing, 52*, 1–19.
De Veirman, M., De Jans, S., Van den Abeele, E., & Hudders, L. (2020). *Unravelling the power of social media influencers: A qualitative study on teenage influencers as commercial content creators on social media*.
De Veirman, M., & Hudders, L. (2020). Disclosing sponsored instagram posts: The role of material connection with the brand and message-sidedness when disclosing covert advertising. *International Journal of Advertising, 39*(1), 94–130.
De Veirman, M., Hudders, L., & Nelson, M. R. (2019). What is influencer marketing and how does it target children? A review and direction for future research. *Frontiers in Psychology, 10*.
Dunlop, S., Freeman, B., & Jones, S. (2016). *Marketing to youth in the digital age: The promotion of unhealthy products and Health promoting behaviours on social media* (Vol. 4, p. 35). Media and Communication.
Evans, N. J., Wojdynski, B. W., & Grubbs Hoy, M. (2019). How sponsorship transparency mitigates negative effects of advertising recognition. *International Journal of Advertising, 38*(3), 364–382.
Fazio, R. H., & Olson, M. A. (2003). Implicit measures in social cognition research: Their meaning and use. *Annual Review of Psychology, 54*(1), 297–327.
Folkvord, F., Anschütz, D. J., Boyland, E., Kelly, B., & Buijzen, M. (2016). Food advertising and eating behavior in children. *Current Opinion in Behavioral Sciences, 9*, 26–31.
Gabrielle, L. (2022). *The origins of the "telling me without telling me" meme*. Available from: https://www.thelist.com/585719/the-origins-of-the-telling-me-without-telling-me-meme/.
Gao, Y., & Lang, A. (2009). *Dimensions of information density and cognitive load*. Chicago: The 59th Annual Conference of the International Communication Association. IL.
Gawronski, B., & Bodenhausen, G. (2006). Associative and propositional processes in evaluation: An integrative review of implicit and explicit attitude change. *Psychological Bulletin, 132*, 692–731.
Geyser, W. (2022). *The state of influencer marketing 2022: Benchmark report*. Influencer Marketing Hub. Available from: https://influencermarketinghub.com/influencer-marketing-benchmark-report/.
Gillespie, E. (2021). *Dirty little secret': The influencers making bank with followers*. Available from: https://www.sbs.com.au/news/the-feed/dirty-little-secret-the-influencers-making-bank-with-followers.
Gupta, P. B., & Gould, S. J. (2007). Recall of products placed as prizes versus commercials in game shows. *Journal of Current Issues and Research in Advertising, 29*(1), 43–53.
Hayes, A. F. (2022). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. Guilford Press.
Hoek, R. W., Rozendaal, E., van Schie, H. T., van Reijmersdal, E. A., & Buijzen, M. (2020). Testing the effectiveness of a disclosure in activating children's advertising literacy in the context of embedded advertising in vlogs. *Frontiers in Psychology, 11*.
John, D. R. (1999). Consumer socialization of children: A retrospective look at twenty-five years of research. *Journal of Consumer Research, 26*(3), 183–213.
Kamleitner, B., & Khair Jyote, A. (2013). How using versus showing interaction between characters and products boosts product placement effectiveness. *International Journal of Advertising, 32*(4), 633–653.
Kelly, B., Bosward, R., & Freeman, B. (2021). Australian children's exposure to, and engagement with, web-based marketing of food and drink brands: Cross-sectional observational study. *Journal of Medical Internet Research, 23*(7), Article e28144.
Klinger, M. R., & Greenwald, A. G. (1994). Preferences need no inferences?: The cognitive basis of unconscious mere exposure effects. In *The heart's eye: Emotional influences in perception and attention* (pp. 67–85). San Diego, CA, US: Academic Press.
La Ferle, C., & Edwards, S. (2006). Product placement: How brands appear on television. *Journal of Advertising, 35*, 65–86.
Lapierre, M. A., Fleming-Milici, F., Rozendaal, E., McAlister, A. R., & Castonguay, J. (2017). The effect of advertising on children and adolescents. *Pediatrics, 140*(Supplement_2), S152–S156.
Lehu, J.-M., & É, B. (2009). Recall of brand placement in movies: Interactions between prominence and plot connection in real conditions of exposure. *Recherche et Applications en Marketing, 24*(1), 7–26.
Linqia. (2021). *The state of influencer marketing 2021: How enterprise marketers view influencer marketing - the need for influencer ROI*.
Lou, C., & Kim, H. K. (2019). Fancying the new rich and famous? Explicating the roles of influencer content, credibility, and parental mediation in adolescents' parasocial relationship, materialism, and purchase intentions. *Frontiers in Psychology, 10*.
Matthes, J., & Naderer, B. (2015). Children's consumption behavior in response to food product placements in movies. *Journal of Consumer Behaviour, 14*(2), 127–136.
Matthes, J., & Naderer, B. (2016). Product placement disclosures: Exploring the moderating effect of placement frequency on brand responses via persuasion knowledge. *International Journal of Advertising, 35*(2), 185–199.
McCarty, J. A. (2004). Product placement: The nature of the practice and potential avenues of inquiry. In L. J. Shrum (Ed.), *The psychology of entertainment media: Blurring the lines between entertainment and persuasion* (pp. 45–61). Mahwah: Lawrence Erlbaum Associates, Inc.
Meléndez-Illanes, L., González-Díaz, C., & Álvarez-Dardet, C. (2022). Advertising of foods and beverages in social media aimed at children: High exposure and low control. *BMC Public Health, 22*(1), 1795.
Morrison, S. (2022). *TikTok is full of shady secret advertisements*. Available from: https://www.vox.com/recode/23197348/tiktok-ad-sponcon-influencers.
Naderer, B., Matthes, J., Marquart, F., & Mayrhofer, M. (2018a). Children's attitudinal and behavioral reactions to product placements: Investigating the role of placement frequency, placement integration, and parental mediation. *International Journal of Advertising, 37*(2), 236–255.
Naderer, B., Matthes, J., & Zeller, P. (2018b). Placing snacks in children's movies: Cognitive, evaluative, and conative effects of product placements with character product interaction. *International Journal of Advertising, 37*(6), 852–870.
Nairn, A., & Fine, C. (2008). Who's messing with my mind? *International Journal of Advertising, 27*(3), 447–470.
Nebenzahl, I. D., & Jaffe, E. D. (1998). Ethical dimensions of advertising executions. *Journal of Business Ethics, 17*(7), 805–815.
Newcom. (2022). *Het Nationale Social Media Onderzoek 2022*.
Nunnally, J. C. (1967). *Psychometric theory*. New York: McGraw-Hill.
Ofcom. (2017). *Children and parents: Media use and attitudes report 2017*. Ofcom.
O'Neil-Hart, C., & Blumenstein, H. (2016). Why YouTube stars are more influential than traditional celebrities. *Think with Google*, 1–2.
Pechmann, C., Levine, L., Loughlin, S., & Leslie, F. (2005). Impulsive and self-conscious: Adolescents' vulnerability to advertising and promotion. *Journal of Public Policy and Marketing, 24*(2), 202–221.
Petty, R. E., & Cacioppo, J. T. (1986). *Communication and persuasion : Central and peripheral routes to attitude change/Richard E. Petty, John T. Cacioppo* (1st ed.). New York: Springer. New York.
Potvin Kent, M., Pauzé, E., Roy, E.-A., de Billy, N., & Czoli, C. (2019). Children and adolescents' exposure to food and beverage marketing in social media apps. *Pediatr Obes, 14*(6), e12508–e.
Raaijmakers, Q. A. W., Hoof, J. T. C., Hart, H., Verbogt, T. F. M. A., & Vollebergh, W. A. M. (2000). Adolescents' midpoint responses on Likert-type scale items: Neutral or missing values? *International Journal of Public Opinion Research, 12*.
Reijmersdal, E. (2009). Brand placement prominence: Good for memory! Bad for attitudes? *Cellular Oncology - CELL ONCOL, 49*, 151–153.
van Reijmersdal, E. A., Boerman, S. C., Buijzen, M., & Rozendaal, E. (2017). This is advertising! Effects of disclosing television brand placement on adolescents. *Journal of Youth and Adolescence, 46*(2), 328–342.
van Reijmersdal, E. A., & van Dam, S. (2020). How age and disclosures of sponsored influencer videos affect adolescents' knowledge of persuasion and persuasion. *Journal of Youth and Adolescence, 49*(7), 1531–1544.
Rozendaal, E., Lapierre, M. A., van Reijmersdal, E. A., & Buijzen, M. (2011). Reconsidering advertising literacy as a defense against advertising effects. *Media Psychology, 14*(4), 333–354.
Rozendaal, E., Slot, N., van Reijmersdal, E. A., & Buijzen, M. (2013). Children's responses to advertising in social games. *Journal of Advertising, 42*(2–3), 142–154.
Rozendaal, E., & van Reijmersdal, E. A. (2017). *Hoe kan de transparantie van reclame in online video's vergroot worden voor minderjarigen? Een literatuuronderzoek naar de effecten van vermeldingen*. University of Amsterdam.
Russell, C. (2002). Investigating the effectiveness of product placements in television shows: The role of modality and plot connection congruence on brand memory and attitude. *Journal of Consumer Research, 29*, 306–318.
Sesar, V., Martinčević, I., & Boguszewicz-Kreft, M. (2022). Relationship between advertising disclosure, influencer credibility and purchase intention. *Journal of Risk and Financial Management, 15*, 276.
Smith, R., Kelly, B., Yeatman, H., & Boyland, E. (2019). Food marketing influences children's attitudes, preferences and consumption: A systematic critical review. *Nutrients, 11*(4), 875.
Sokolova, K., & Kefi, H. (2020). Instagram and YouTube bloggers promote it, why should I buy? How credibility and parasocial interaction influence purchase intentions. *Journal of Retailing and Consumer Services, 53*, Article 101742.
Spielvogel, I., Matthes, J., Naderer, B., & Karsay, K. (2018). A treat for the eyes. An eye-tracking study on children's attention to unhealthy and healthy food cues in media content. *Appetite, 125*, 63–71.
Tulving, E., Schacter, D. L., & Stark, H. A. (1982). Priming effects in word-fragment completion are independent of recognition memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 8*(4), 336–342.
Valenzuela, L., Martinez-Troncoso, C., & Yanez-Wieland, F. (2015). Influence of placement on explicit and implicit memory of college students. *Comunicar, 22*, 169–176.
van der Bend, D. L. M., Beunke, T. A., Shrewsbury, V. A., Bucher, T., & Kleef, E. (2022a). *My feed is what I eat? A qualitative analysis of adolescents' awareness and appreciation of food marketing on social media*. Public Health Nutr (submitted for publication).
van der Bend, D. L. M., Jakstas, T., van Kleef, E., Shrewsbury, V. A., & Bucher, T. (2022b). Adolescents' exposure to and evaluation of food promotions on social media: A multi-method approach. *International Journal of Behavioral Nutrition and Physical Activity, 19*(1), 74.
van der Bend, D. L. M., Jakstas, T., van Kleef, E., Shrewsbury, V. A., & Bucher, T. (2022c). Making sense of adolescent-targeted social media food marketing: A qualitative study of expert views on key definitions, priorities and challenges. *Appetite, 168*, Article 105691.
Van Der Goot, M. J., Van Reijmersdal, E. A., & Zandbergen, S. K. P. (2021). Sponsorship disclosures in online sponsored content: Practitioners' considerations. *Journal of Media Ethics, 36*(3), 154–169.

Vanwesenbeeck, I., Ponnet, K., & Walrave, M. (2017). Young adolescents' advertising literacy and purchase intention in social network games: Influence of perspective taking and need for cognition. *Journal of Consumer Behaviour, 16*(1), 23–33.

Verhellen, Y., Oates, C., De Pelsmacker, P., & Dens, N. (2014). Children's responses to traditional versus hybrid advertising formats: The moderating role of persuasion knowledge. *Journal of Consumer Policy, 37*(2), 235–255.

Weismueller, J., Harrigan, P., Wang, S., & Soutar, G. N. (2020). Influencer endorsements: How advertising disclosure and source credibility affect consumer purchase intention on social media. *Australasian Marketing Journal, 28*(4), 160–170.

World Health Organization. (2016 2016). *Regional office for E. Tackling food marketing to children in a digital world: trans-disciplinary perspectives: children's rights, evidence of impact, methodological challenges, regulatory options and policy implications for the WHO European region*. Copenhagen: World Health Organization. Regional Office for Europe.

Wright, P., Friestad, M., & Boush, D. M. (2005). The development of marketplace persuasion knowledge in children, adolescents, and young adults. *Journal of Public Policy and Marketing, 24*(2), 222–233.

Yang, M., & Roskos-Ewoldsen, D. R. (2007). The effectiveness of brand placements in the movies: Levels of placements, explicit and implicit memory, and brand-choice behavior. *Journal of Communication, 57*(3), 469–489.